

Cheng-Yeh Chen

Ph.D. Student · BCI · NeuroAI · Georgia Tech ECE

Aligning brains and transformers – to interpret AI and decode the mind.

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COMPETITIVE BENCHMARKS SOTA on open, verifiable leaderboards

1st / 39 BRAIN-T0-TEXT 2024 Intracortical speech BCI WER 4.57% — world SOTA, held since June 2026.	1st / 23 ID 2nd / 20 OOD ALGONAUTS 2025 · POST-CHALLENGE Multimodal fMRI encoding Pearson r = 0.335 across ~1,000 parcels on 80h of film.	5th / 53 NEURIPS 2025 LIBRIBRAIN MEG speech detection F1 = 0.907 (1st 0.917) on 306-ch MEG.
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EDUCATION

Ph.D. Student, ECE

Georgia Institute of Technology
2023 – present
Advisor: Raghupathy Sivakumar

M.S., Communication Eng.

National Taiwan University
2019 – 2021
Advisor: Hung-Yun Hsieh

B.S., Electrical Eng.

National Taiwan University
2015 – 2019

SKILLS

METHODS

Transformer probing · brain encoding / decoding · fMRI / EEG / MEG / iEEG

STACK

PyTorch · Hugging Face · nilearn

LANGUAGES

Python · C / C++ · MATLAB · Bash

TEACHING

Probability & Statistics

TA · NTU · 2020

Computer Programming (C++)

TA · NTU · 2019 – 2020

SERVICE

Reviewer — NeurIPS, IEEE ICC, IEEE TVT, IEEE Access

Research Focus

I measure **where and how brains and AI models align**, then run that alignment both ways — **encoding** brain activity from language / vision / speech models, and **decoding** perception directly from fMRI, MEG, EEG, and iEEG.

Publications & Projects

ICLR 2026 · TOP 7% REVIEW SCORE **The Mind’s Transformer: Computational Neuroanatomy of LLM–Brain Alignment**

First systematic, anatomical, and intra-block analysis of where and how LLMs and the brain align — the anchor of my research program.

SELECTED PROJECTS · FOUR MANUSCRIPTS UNDER REVIEW

- **EEG → image** — new SOTA, >95% top-1 (~20-pt jump), ~10× trial efficiency.
- **EEG → text** — new SOTA, ~30%+ over prior under a leakage-free protocol.
- **fMRI → image scaling** — retrieval near-perfect (>95%) on a 100K+ pool (prior <50%).
- **ViT via cortex** — cortical topography traces ViT internals (staged circuit; attention/FFN division of labor).

ECCV 2026 LoViF WORKSHOP **Frozen Decoder, Better Bits: Per-Image Latent Rate-Distortion Optimization for Ultra-Low Bitrate Compression**

#1 / 26 on the LoViF ultra-low bitrate compression dev leaderboard (composite 72.643 at 0.008 BPP), **2nd** in the human-judged final — best fidelity in the field at a competitive perceptual score.

Research Experience

Georgia Tech · GNAN Lab

Aug 2023 – Present
Graduate Research Assistant · NeuroAI · Advisor: Prof. Raghupathy Sivakumar

- Measured where and how LLMs align with the brain — the first anatomical, intra-block LLM–fMRI study (*MindTransformer*, ICLR 2026, Top 7% reviews).
- Built multimodal brain encoders ranked **#1 / 23** on the live Algonauts 2025 fMRI benchmark (post-challenge, In-Distribution).
- Led *CoDec*, holding **#1 / 39** on the live Brain-to-Text 2024 board since June 2026 — world-SOTA intracortical attempted-speech decoding at 4.57% WER.
- Set new state of the art decoding images, text, and speech from fMRI / MEG / EEG (four manuscripts under review).